How performance influences the result of the game. Numbers and Nerves

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Object of research: Dataset of CS2 pro matches (05.10.2023 - 31.05.2026)

MythBusting

1. Simple, unchangeable game conditions
2. Win the first map - win the series
3. More damage = win
4. You need a superstar to win
5. Fraggers (High ADR) are the most valuable players

Bonus question

1. Top performing nations and where is Ukraine placed there

Data & Preparation

Before

- 23 888 rows
- Matches, maps and players are all mixed up

After

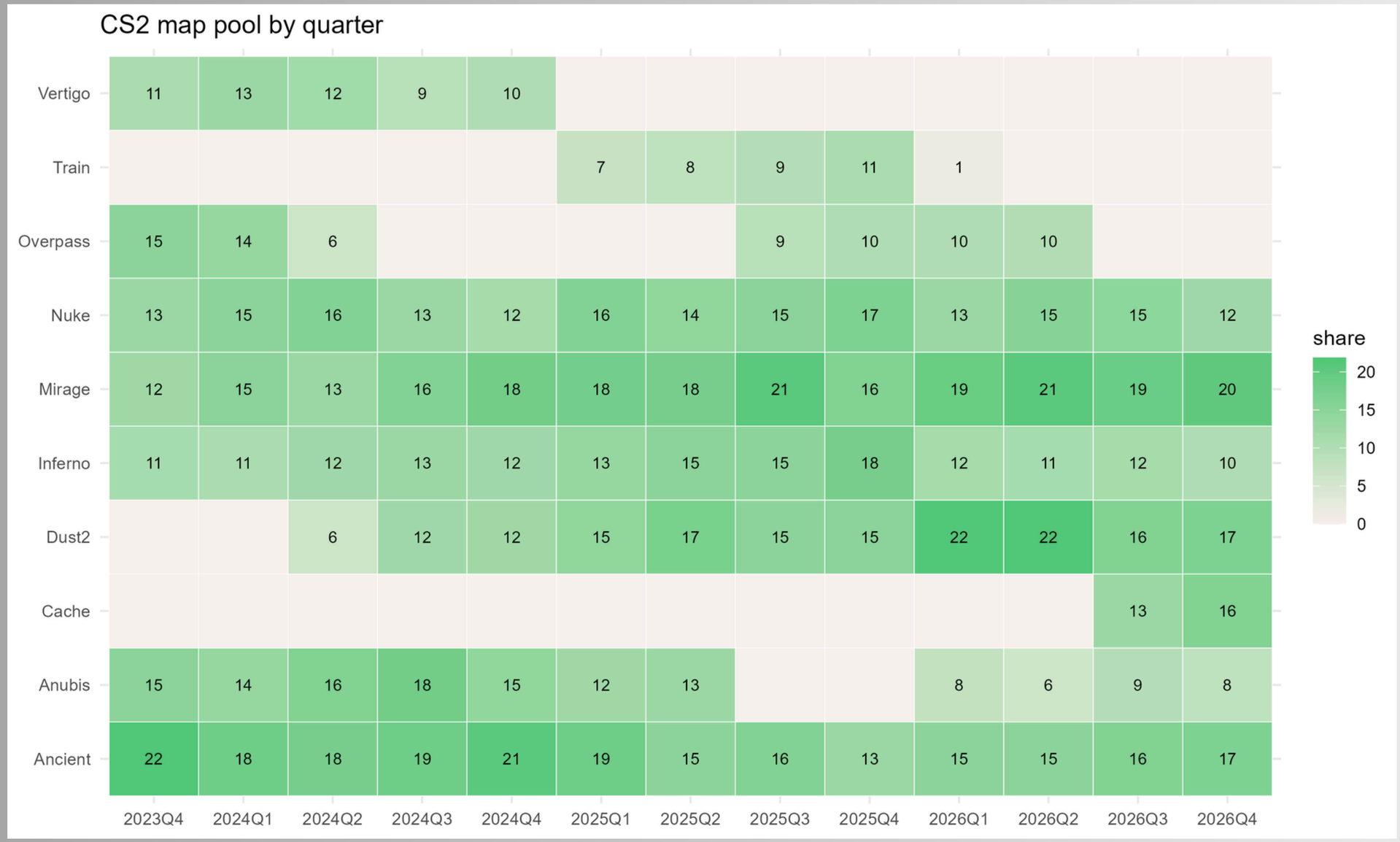
- 9888 matches
- 11555 maps
- 110692 player-map records
- 22839 team-map records
- 1216 players
- 687 teams

source: https://www.kaggle.com/datasets/ektarr/counter-strike-pro-matches?select=cs2_tier1_games.csv

Cybersport vocabulary

- **ADR - average damage per round**
- **KAST - % of rounds where player got a Kill, Assist, Survived or was Traded**
- **Tier - level of the tournament (Tier 1 is the biggest ones)**
- **CV(coefficient of variation) - how uneven is ADR inside the team (in %)**
- **Bo3 - standard format for esports matches (particularly CS2), in which the first team to win 2 maps wins.**
- **Frag = 1 kill(point)**

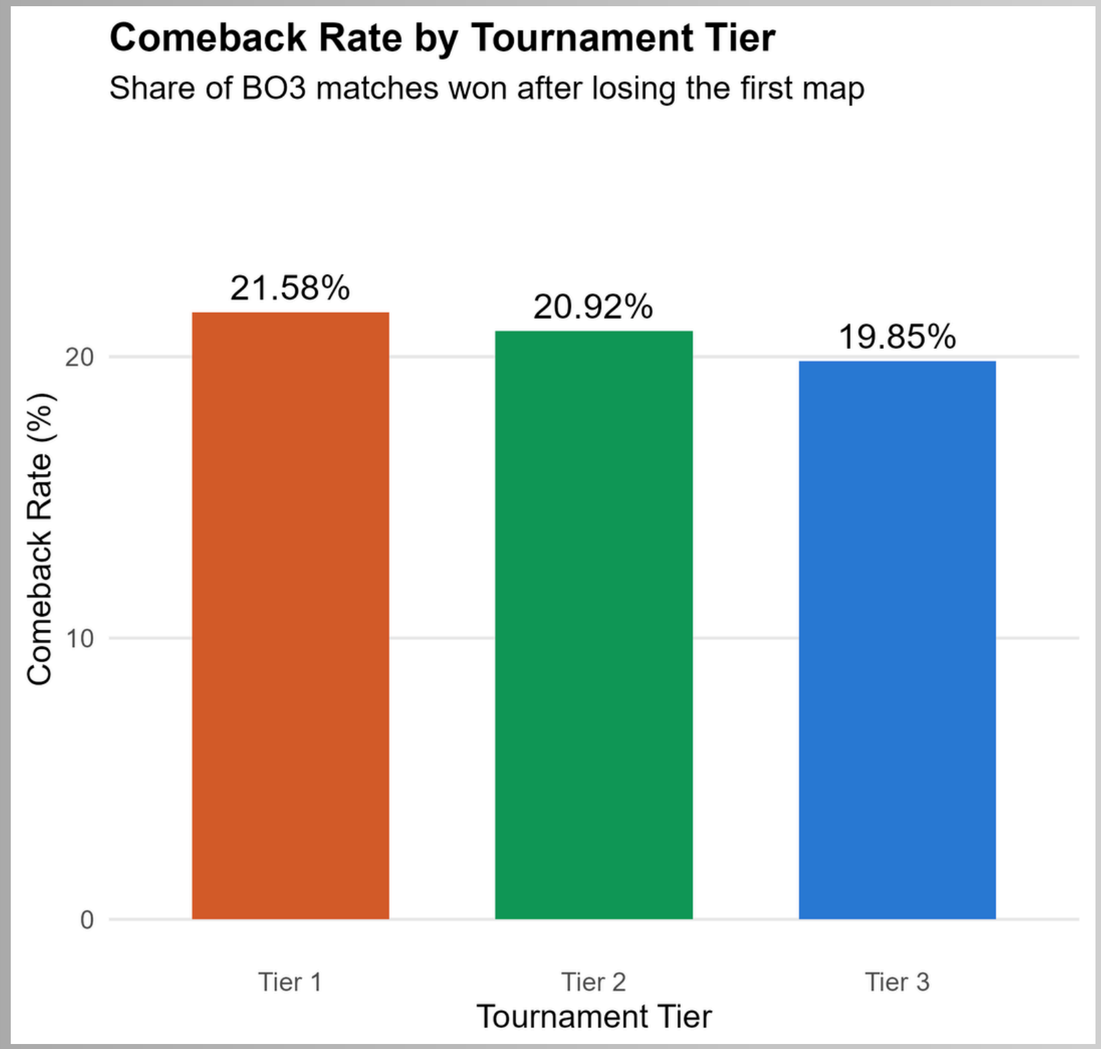
The arena keeps changing

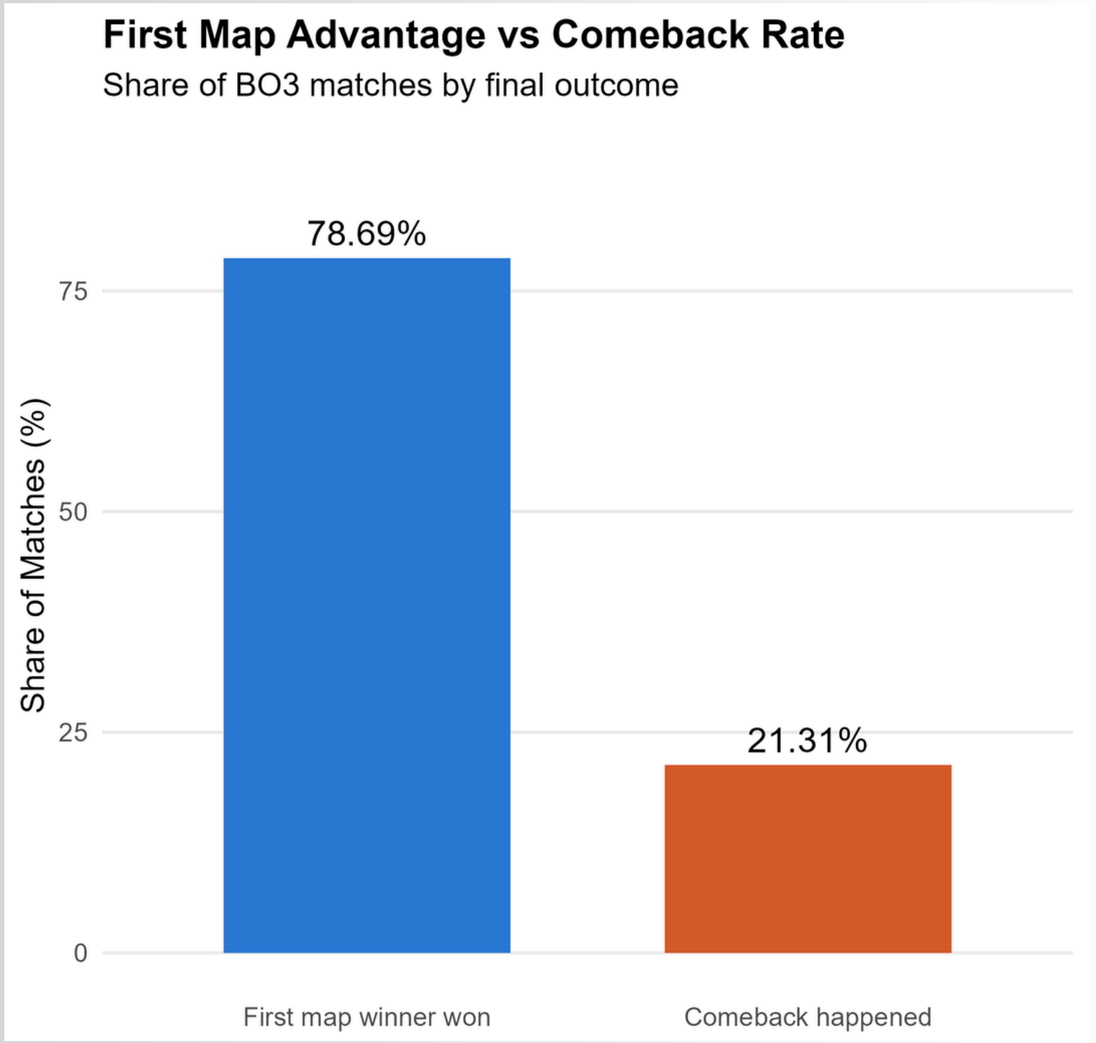


Verdict: myth busted.

Teams compete in constantly changing conditions.

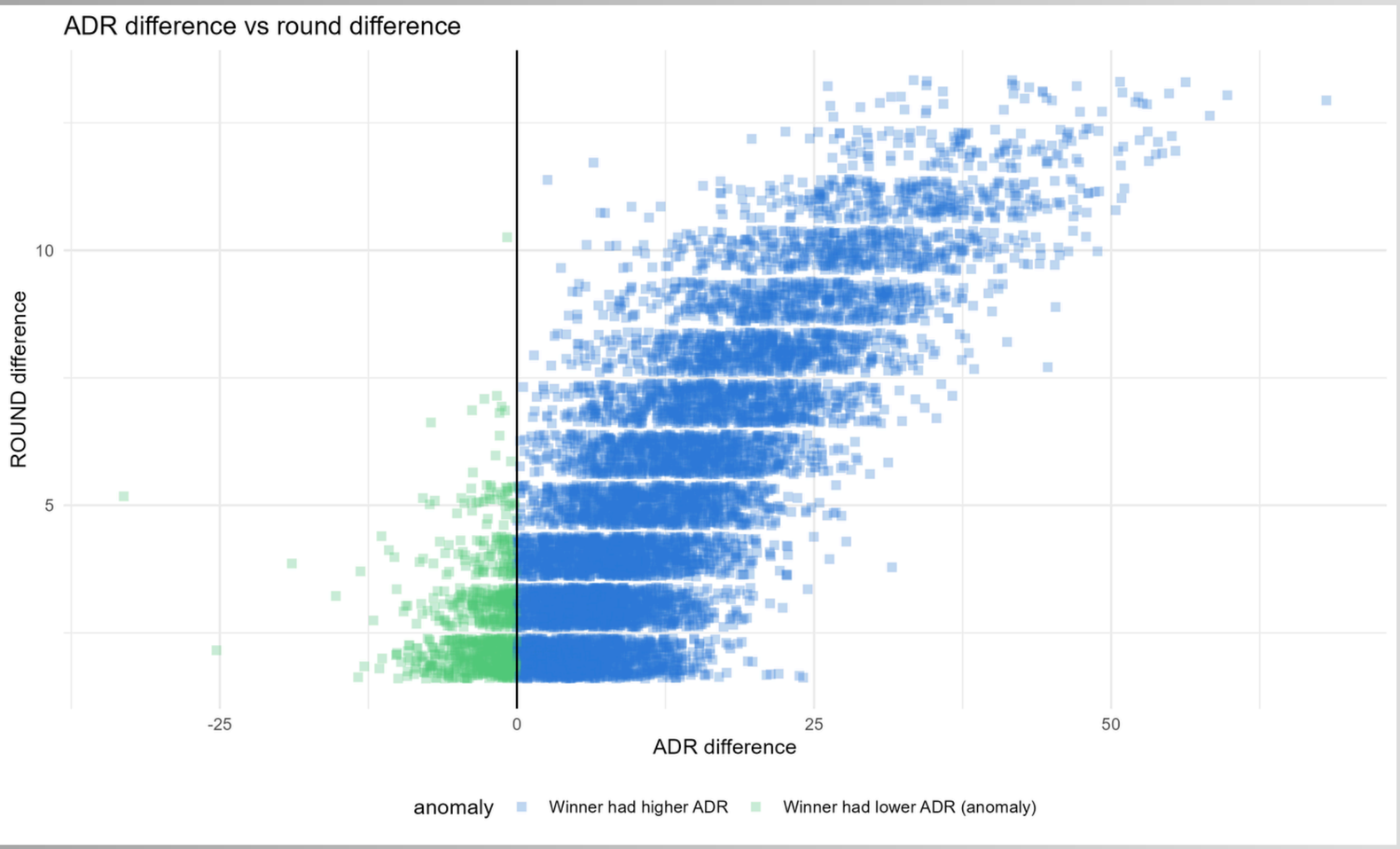
Win the first map = win the series?

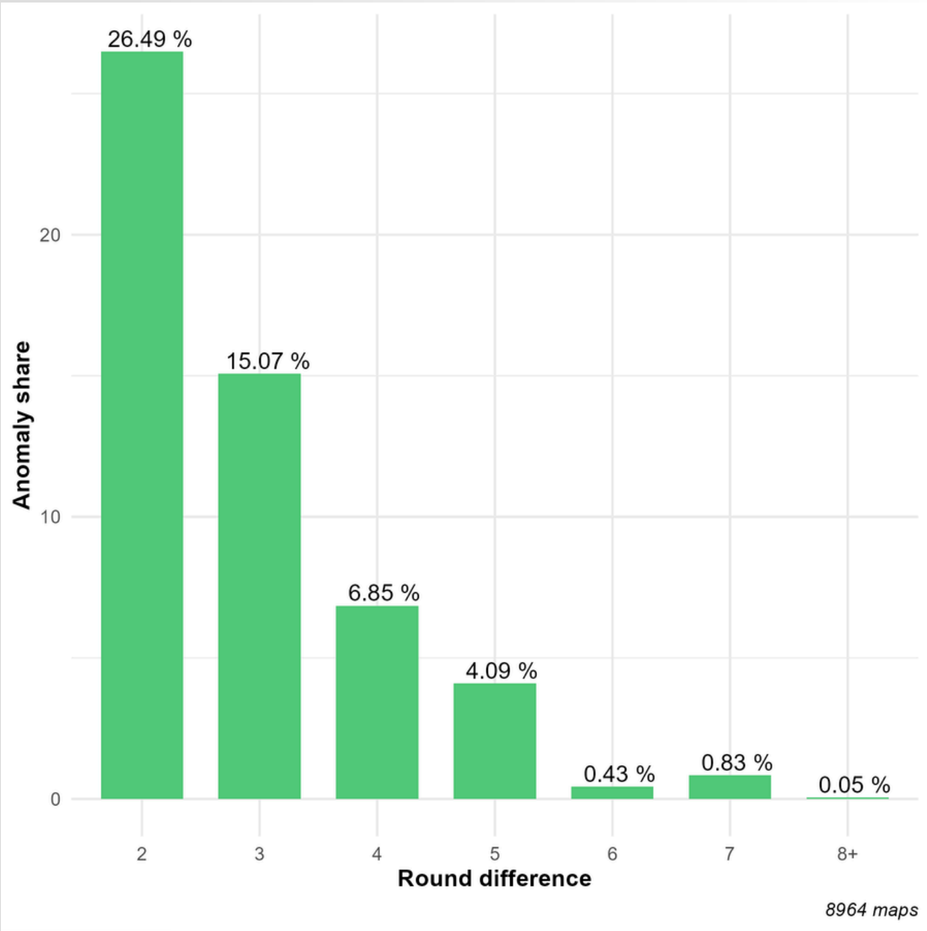




Verdict: half-truth. The first map matters, but it doesn’t decide the series.

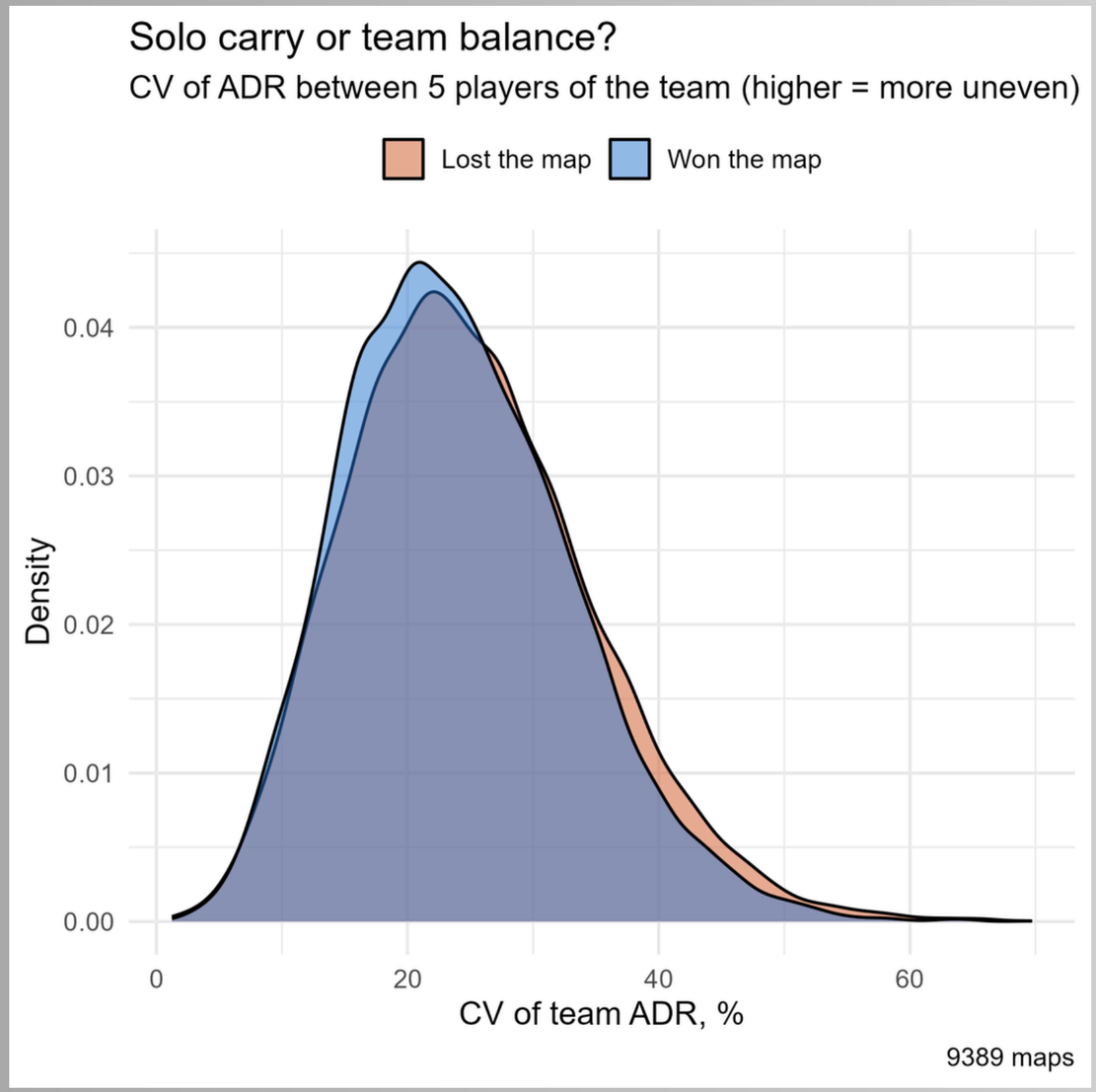
More damage does not guarantee a win

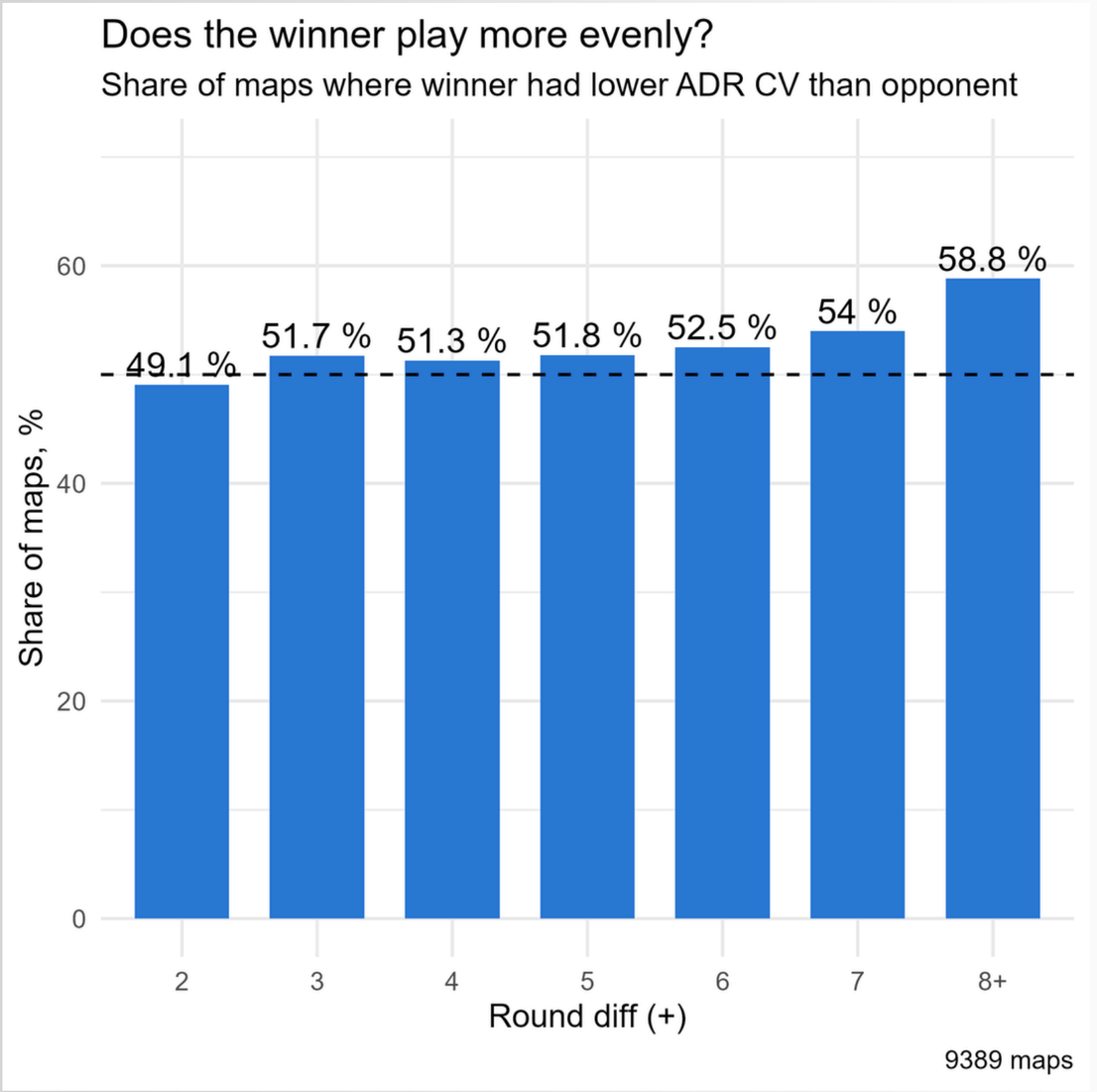




Verdict: half-truth. Damage explains crushing victories, not close games.

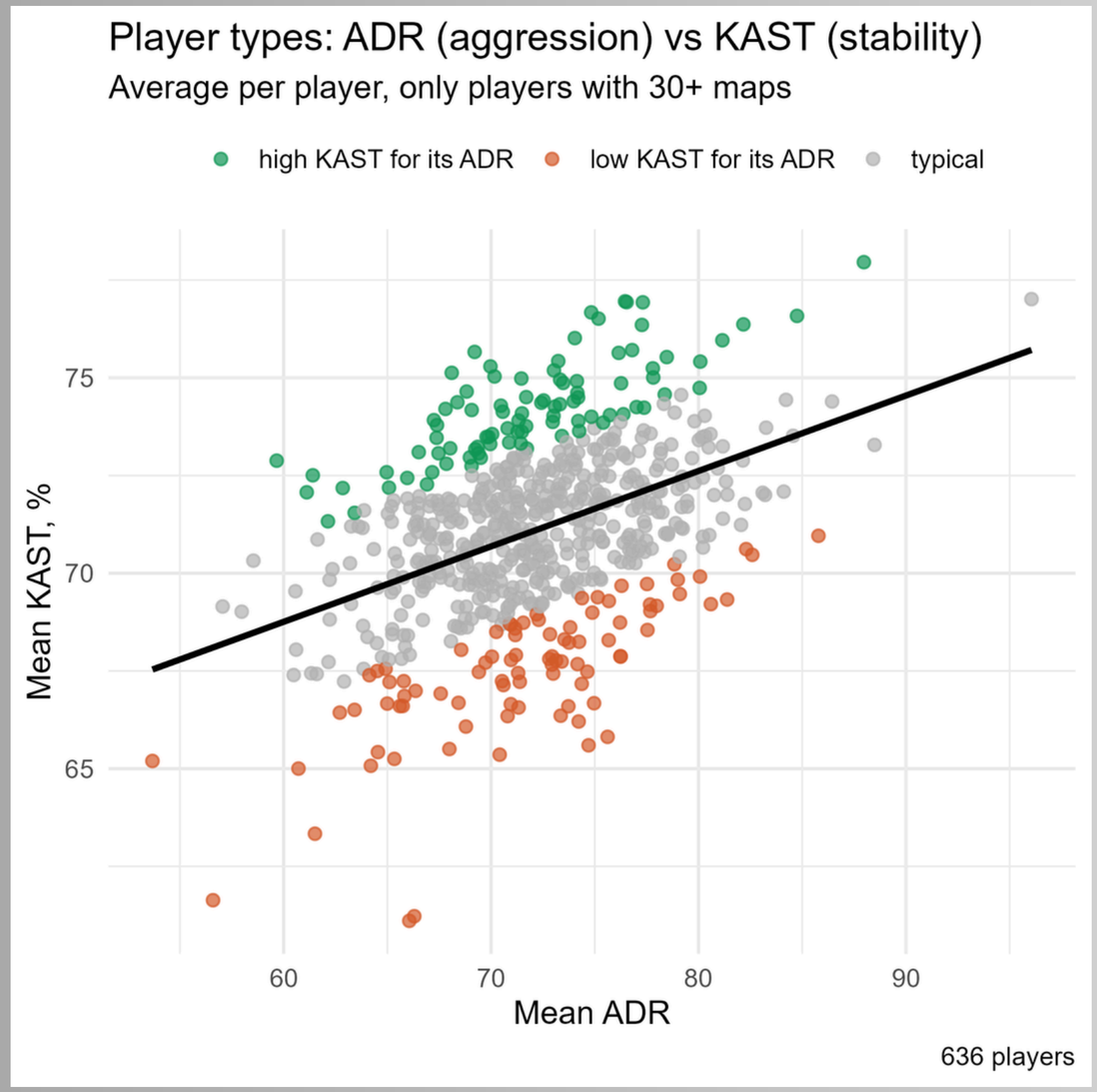
Solo carry or a team?

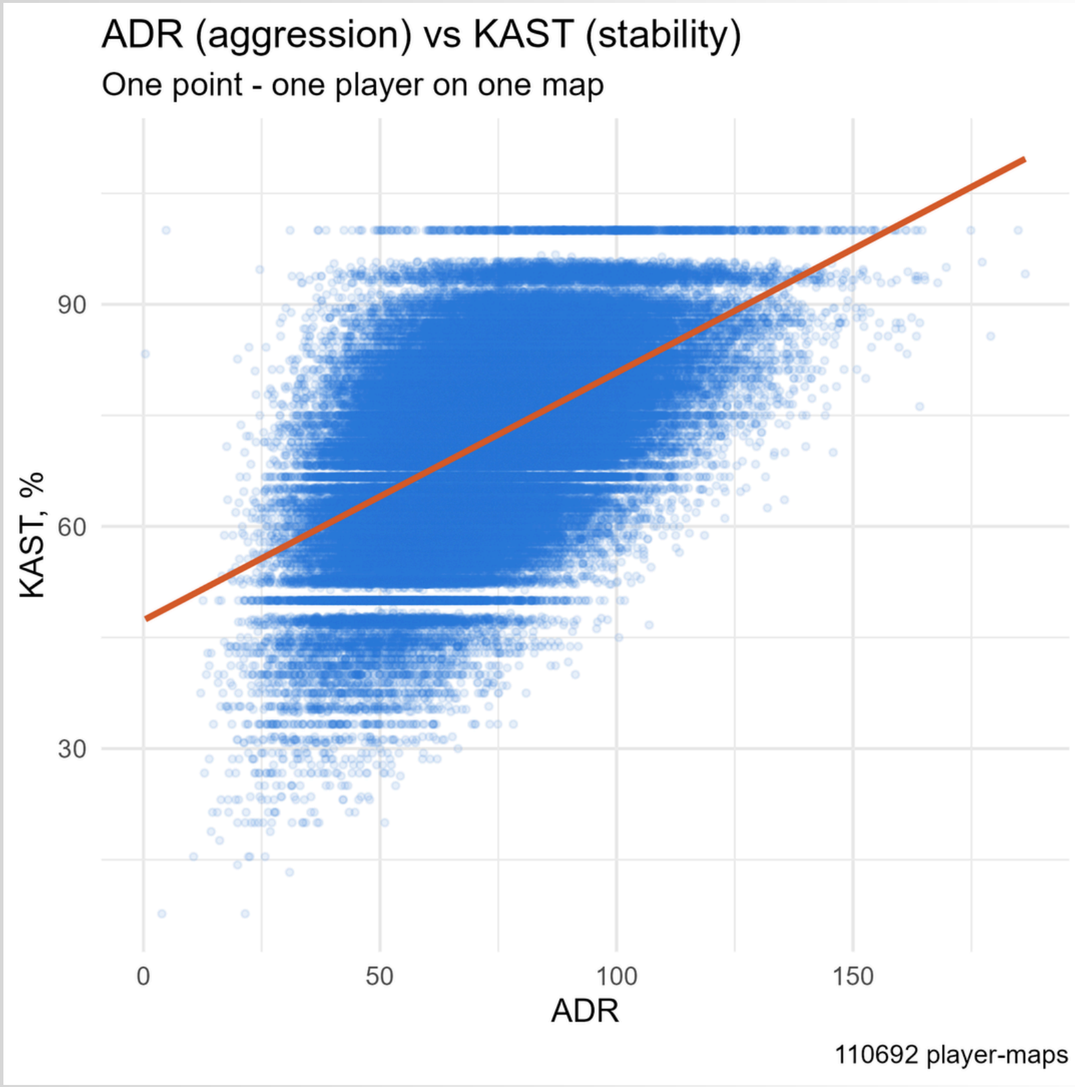




Verdict: myth busted. Neither a superstar nor perfect balance decides the game.

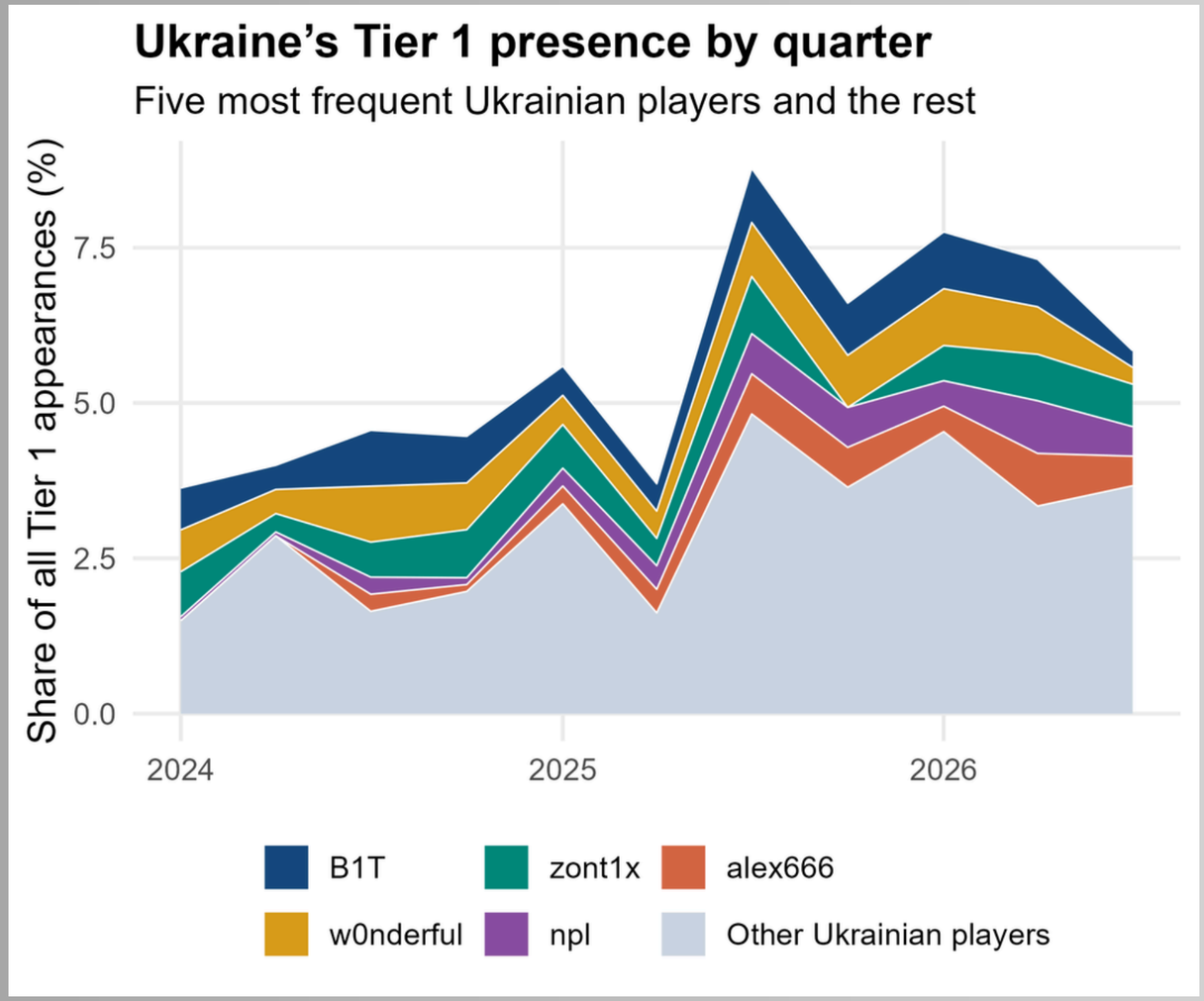
ADR vs KAST

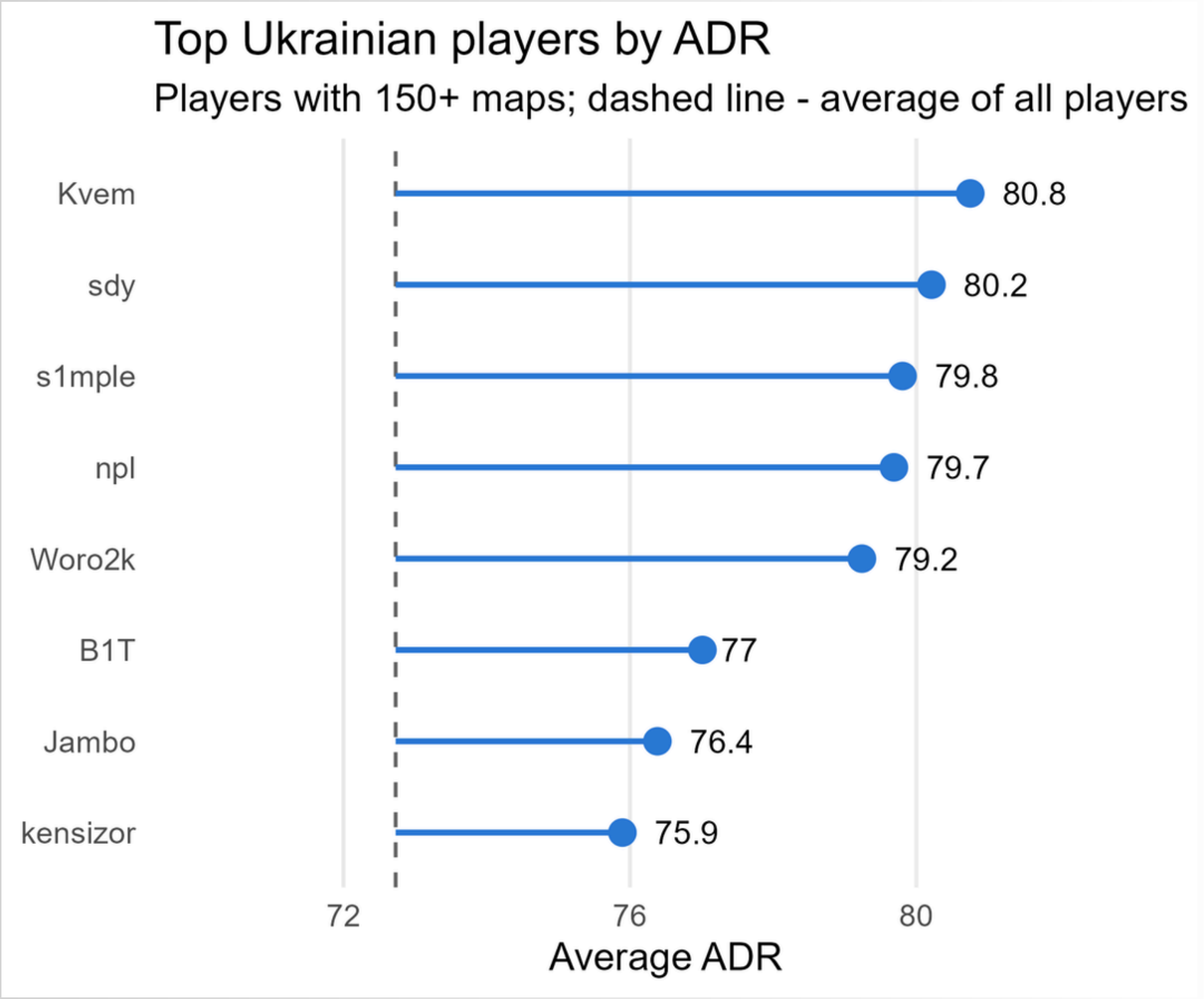




Verdict: half-truth. High ADR also means high KAST (because it includes kills and assists) but you do not require high ADR to perform great on KAST.

How Ukrainians perform





Key findings

- **Game conditions change:** only 4 core maps, other rotate
- **Win map 1 = Win the series:** Only in 78.7% of matches
- **More damage = win:** True, but not in a close games
- **You need a superstar:** neither a superstar nor balance decides; only extreme imbalance hurts
- **Fraggers are most valuable:** consistency (KAST) is linked to winning more than damage

Bonus: Ukrainian players are above average, with a growing tier-1 presence (4.1% to 6.7%).